

Model-Results-Metrics

July 13, 2026

1 Model Results Metrics

1.1 Accuracy

Accuracy measures the percentage of correctly classified samples and is calculated as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Key:

- **TP:** True Positives
- **FN:** False Negatives
- **FP:** False Positives
- **TN:** True Negatives

Implementation: Accuracy is the standard metric for model performance and is widely used for all machine learning models. However, other metrics, such as precision, should be reported alongside accuracy for an in-depth analysis of model performance.

1.2 Precision

Precision measures the model's accuracy of positive predictions and is calculated through the following:

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}} = \frac{TP}{TP + FP}$$

Implementation: For a BBB permeability model, high precision is crucial because it determines how reliable the model's positive (BBB+) labels are.

1.3 Recall

Recall measures how many of the total positive samples the model correctly identified and is calculated as follows:

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}} = \frac{TP}{TP + FN}$$

Implementation: Recall is an important metric for a BBB permeability model because classifying a compound as BBB- when it really is BBB+ can lead to the discarding of a potentially valuable drug.

1.4 F1 Score

F1 score is the harmonic mean of precision and recall, and is calculated through the following:

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP + FP + FN}$$

Implementation: Having a high F1 score is crucial in a BBB permeability model because it ensures that both the model’s precision and recall are good.

1.5 Specificity

Specificity measures how many of the total negative samples the model correctly identified and is calculated as follows:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

Implementation: Specificity is crucial in a BBB permeability model because incorrectly identifying a BBB- compound as BBB+ can lead to unsuccessful and time-consuming drug testing trials.

1.6 AUC-ROC

AUC-ROC measures the area under the ROC (Receiver Operating Characteristic) curve. A ROC curve plots the model’s true positive and false positive rates over all classification thresholds. TPR (true positive rate) and FPR (false positive rate) are calculated as follows:

$$\text{True Positive Rate} = \text{Sensitivity} = \text{Recall} = \frac{TP}{TP + FN}$$

$$\text{False Positive Rate} = 1 - \text{Specificity} = \frac{FP}{FP + TN}$$

Implementation: AUC-ROC represents how well the model is able to differentiate between positive and negative cases. Having a high AUC-ROC, therefore, is important in a BBB permeability model.